

Stat 431: Statistical Analysis

Instructor: Benjamin Ridenhour

Fall 2024

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Office Hours: TR 4:00–5:00pm (or by appt.)

Office: Brink Hall 327

Web: <https://bit.ly/2scoPgS>

Class Hours: MWF 9:30–10:20am

Class Room: TLC 029

Course TA: Jacob Hofer (jhofer@uidaho.edu)

Course Description

Statistics is the science of collecting, organizing and interpreting numerical data. Statistical literacy allows people to understand and make rational decisions based on the analysis of data, in research contexts and in daily life. This course will focus on building an understanding of many statistical concepts and commonly used statistical methods. A sample of the procedures that we will cover include: multiple regression, multiple hypothesis testing, various experimental designs, analysis of variance, and analysis of covariance, contingency tables, and χ^2 -tests. We hope to develop each student's understanding of the role of statistics in research and in daily life. In particular, there is a focus on developing the ability to apply appropriate statistical methods to data, interpreting results, and communicating findings to a broader audience. Lastly, we will use R and RStudio for performing analyses in order to give students exposure to this commonly used statistical programming language.

The structure of the class is as follows. Homework will be assigned for each chapter we discuss. I expect you to have read the appropriate section materials *prior to* each class period. Class periods will be used for lecturing, discussion of the homework set, and strategies for solving problems. *You should be prepared to participate in discussions!* Student assessment is based around completion of the homework, 3 exams, and a final (comprehensive) exam. Dates of the 3 exams will be determined based on progress of the class and announced at least a week in advance (they will roughly occur after every 4 weeks).

Required Materials

- *An Introduction to Statistical Methods & Data Analysis*. 7th edn. by RL Ott and M Longnecker. Cengage Learning: Boston.

- We will use R and RStudio, therefore you will need to have a functioning version of these softwares. Follow the directions at <https://posit.co/download/rstudio-desktop/> to install the appropriate versions. If you do not wish to directly install these programs on your personal computer, you may also use the cloud version of RStudio available at <https://posit.cloud>. Note that the cloud version is limited in its computing time; to have unlimited computing time you must pay for a subscription.

Prerequisites/Corequisites

Prerequisites: Stat 251

Preferred: Stat 301 or Stat 416

Topics

1. Linear Regression and Correlation (Chapter 11)
2. Multiple Regression and the General Linear Model (Chapter 12)
3. Further Regression Topics [model building] (Chapter 13)
4. Inferences About More Than Two Population Central Values (Chapter 8)
5. Multiple Comparisons (Chapter 9)
6. ANOVA and ANCOVA using various designs (Chapters 14–18)

Course Objectives/Learning Outcomes

Successful students will:

1. Develop an understanding of the role of statistics in scientific research and develop the ability to apply appropriate statistical methods to their data.
2. Understand multiple linear regression and demonstrate an ability to interpret and assess the validity of the resulting model.
3. Distinguish between various common experimental designs and demonstrate an ability to interpret and assess the validity of these methods.
4. Use other common statistical models such as ANCOVA and χ^2 -tests for categorical data.

Final Exam Date

9 December (Monday), 8:00–10:00am

Grading Policy

A typical grading scale will be used (i.e., $\geq 93\% = A$, $(93,90\%] = A-$, $(90,87\%] = B+$, ...). I reserve the right to curve the scale dependent on overall class scores at the end of the semester. Any curve will only ever make it easier to obtain a certain letter grade. The grade will count the assessments using the following proportions:

- 30% of your grade will be determined by completing homework.
- 45% of your grade will be determined by 3 equally-weighted exams.
- 25% of your grade will be determined by the final exam.